# An efficient ECG signals denoising technique based on the combination of particle swarm optimisation and wavelet transform

# Update:

- Wavelet Basis Function Plotting:
  - Implemented and visualized wavelet basis functions $g_{j,k}(n)$by varying both scale $(j)$ and shift $(k)$
  - Demonstrated the effects of scaling and shifting on wavelet basis functions using MATLAB plot.
- Evaluation of Noise Removal Capability:
  - Analyzed how varying noise levels impact the quality of reconstructed ECG signals
  - Compared noisy and denoised ECG signals across increasing noise intensities to determine the maximum noise level at which the original ECG morphology can be preserved
- Reconstruction and Testing on Multiple Datasets:
  - Tested the wavelet-based denoising and reconstruction process on different ECG datasets.

# Wavelet Basis Function Plotting:

› *DWT is defines as :*

$$C(a,b) = \sum_{n \in Z} x(n)\ g_{j,k}(n)$$

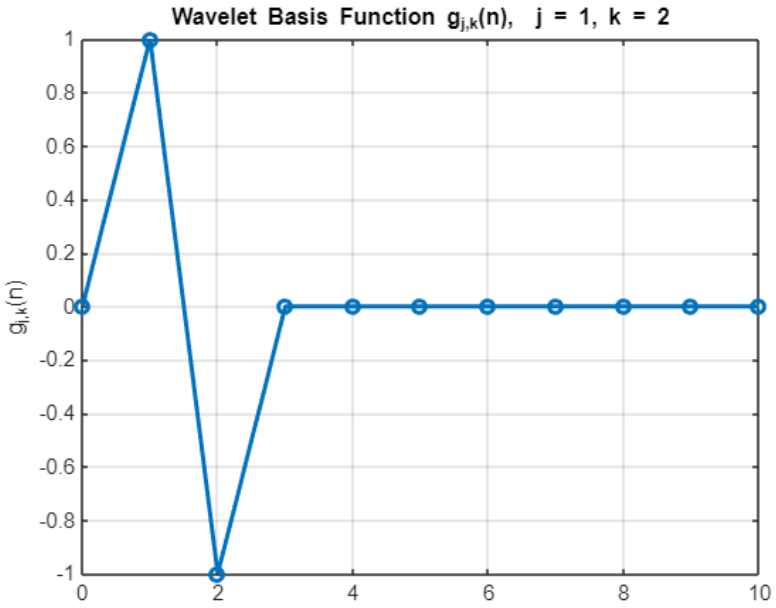
- C(a,b): Wavelet coefficient representing signal content at scale $a$and location $b$
- x(n): Input signal (ECG signal)
- $g_{j,k}(n)$ :Wavelet basis function used for analysis

› Wavelet Basis Function:

$$g_{j,k}(n) = 2^{j/2} g\left(2^j n - k\right)$$

- $g_{j,k}(n)$ :Wavelet basis function at scale $j$and shift $k$
- $g(n)$ :Mother wavelet
- j: Scale parameter controlling time–frequency resolution
- k: Shift parameter controlling time localization
- $n$: Discrete time index

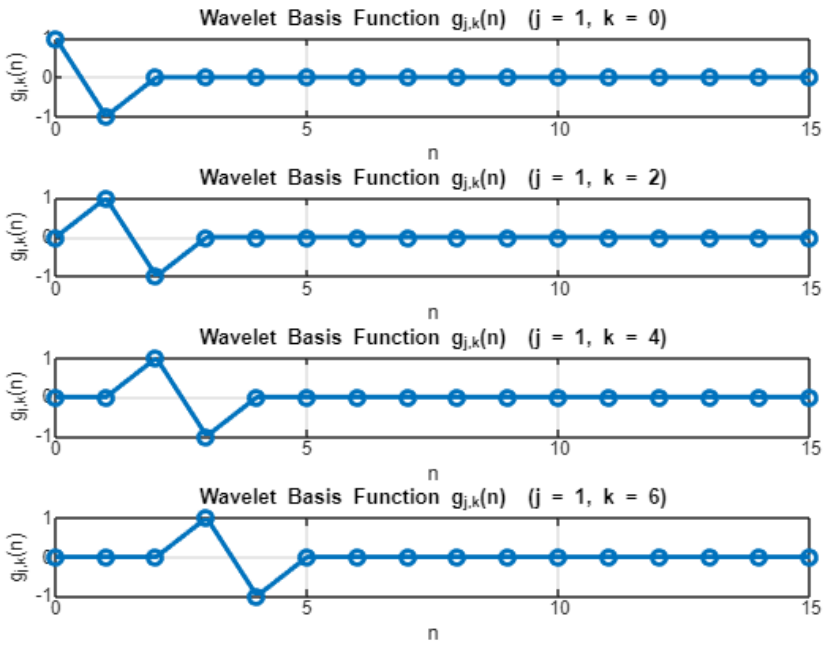
# PLOTS: (Basis Function plotting)



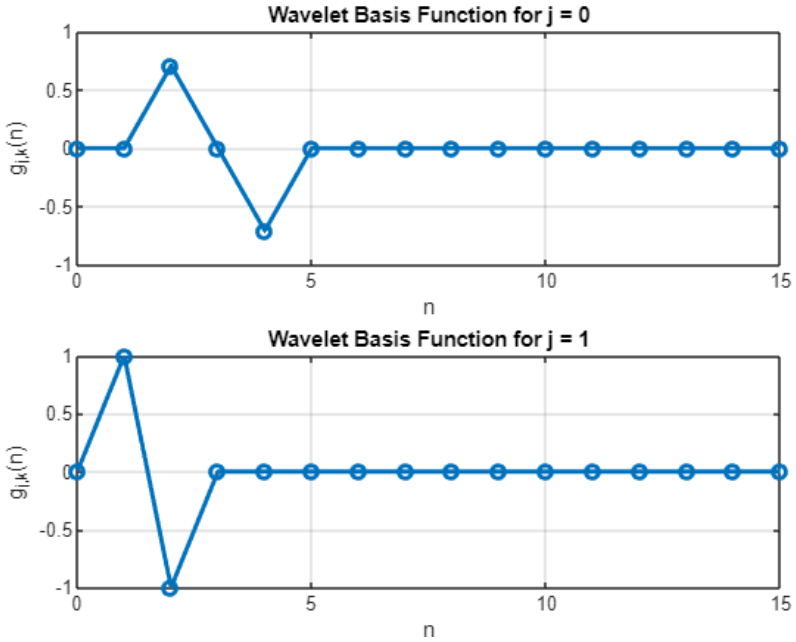

- The non-zero samples indicate where the **scaled and shifted mother wavelet** is active; zeros elsewhere show time localization

- Changing $k$would shift this shape along the time axis, while changing $j$would compress or expand it in time.

# PLOTS: (Shifting and Scaling)




- Each subplot shows the *same wavelet shape* shifted along the time axis for different values of $k$.
- This demonstrates the time localization property of wavelets changing $k$ moves the basis function without altering its shape.



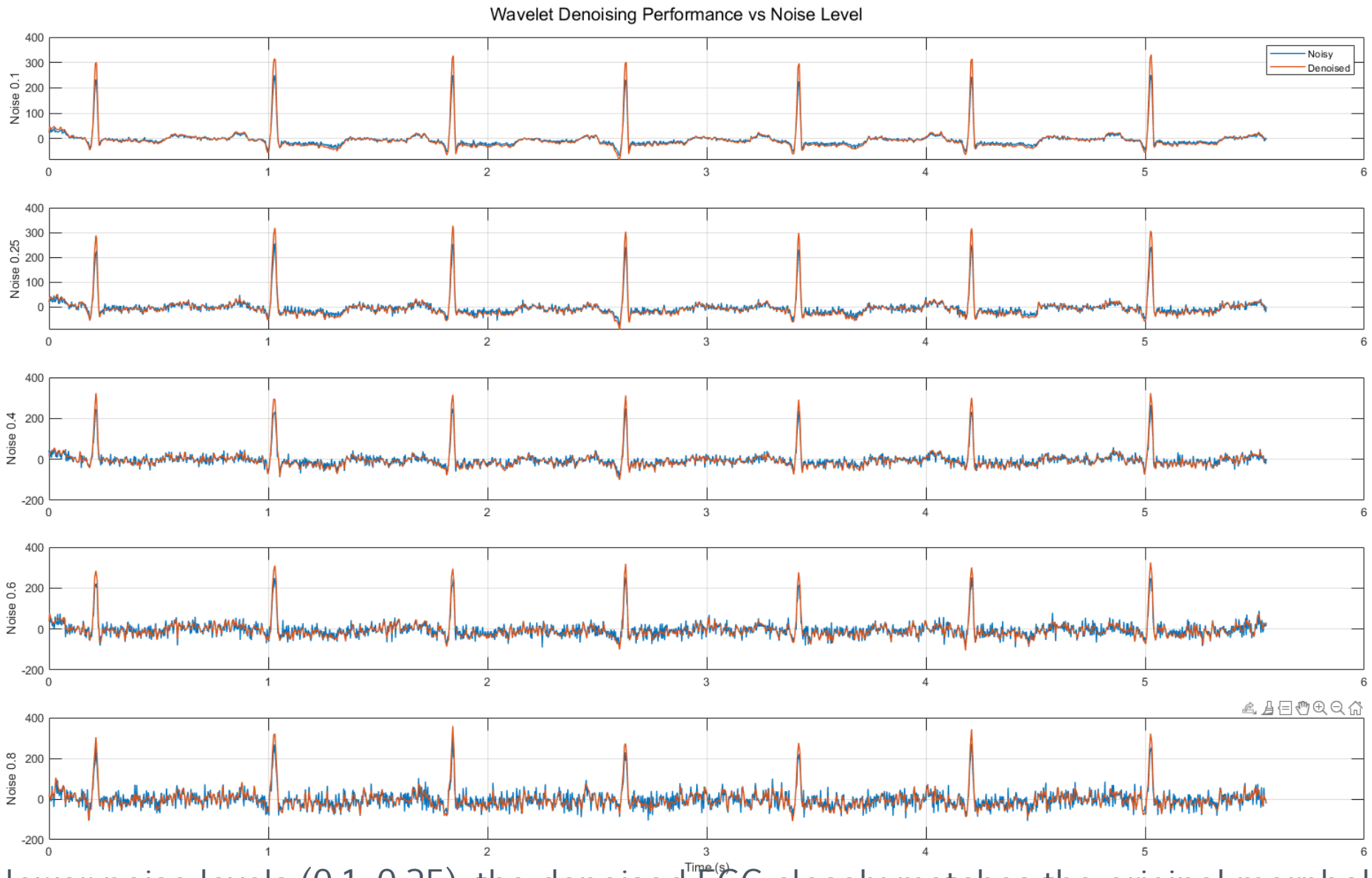

- As $j$ increases, the wavelet becomes more compressed in time.
- This plot demonstrates how changing the scale parameter $j$ alters the width of the wavelet basis function.

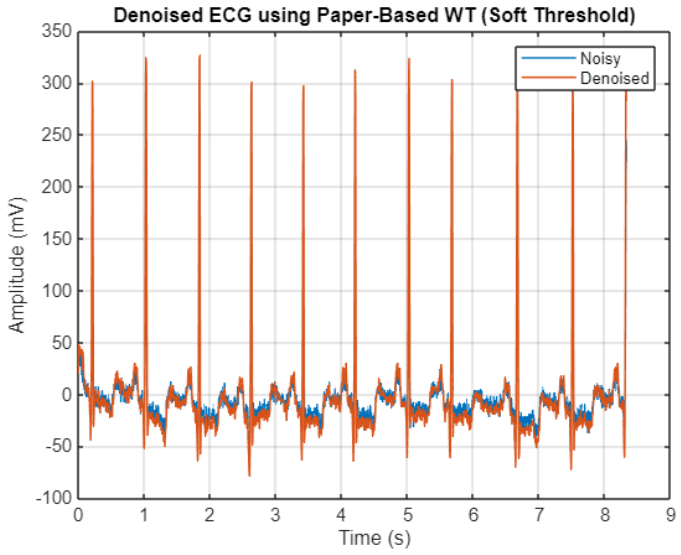
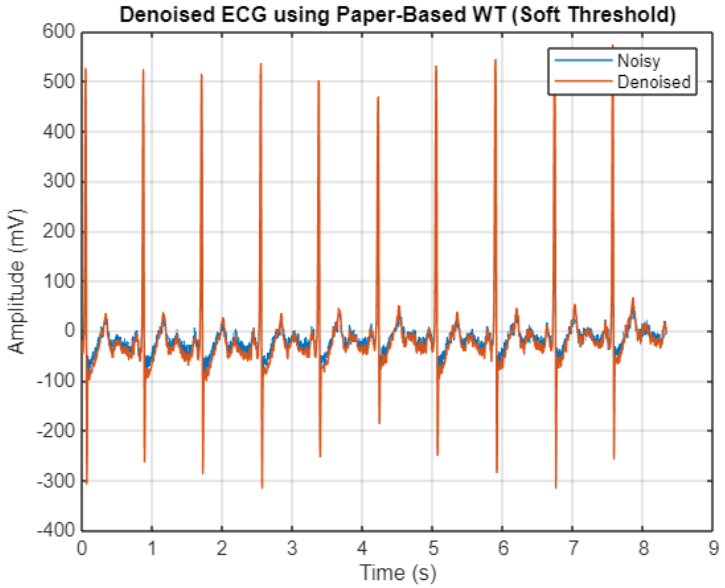
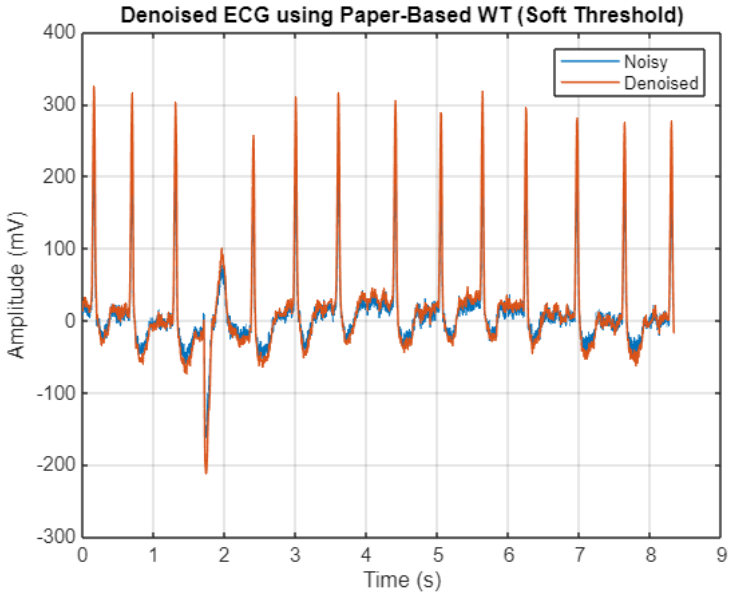
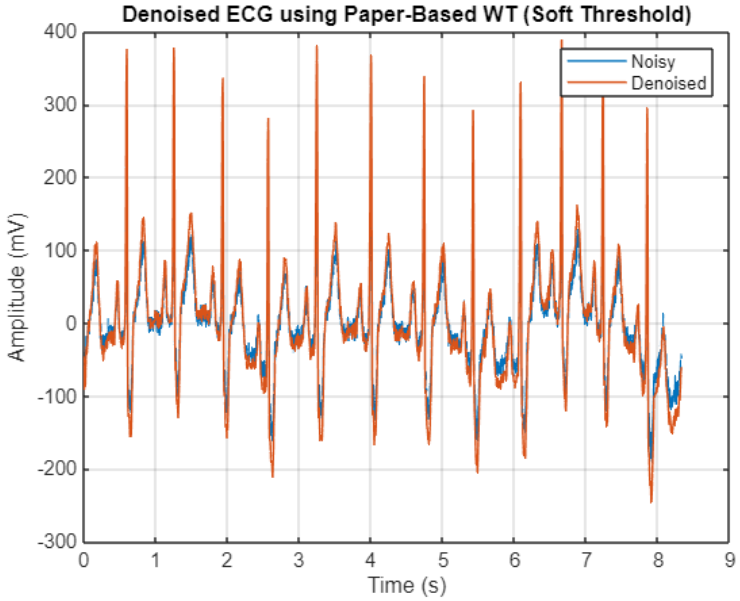
# Evaluation of Noise Removal Capability Using Wavelet Denoising

- Clean ECG signal was corrupted with Additive White Gaussian Noise (AWGN) at different noise levels:

$$K = 0.1,\ 0.25,\ 0.4,\ 0.6,\ 0.8$$

- The noisy ECG signal was decomposed using a multi-level discrete wavelet transform (DWT).
- Soft thresholding was applied to the detail coefficients to suppress noise.
- The ECG signal was reconstructed using inverse wavelet transform.




- At lower noise levels (0.1–0.25), the denoised ECG closely matches the original morphology, including clear QRS complexes.
- As the noise level increases, the noisy signal becomes highly distorted, but the wavelet denoising method still preserves major ECG features.

# Reconstruction and Testing on Multiple Datasets: